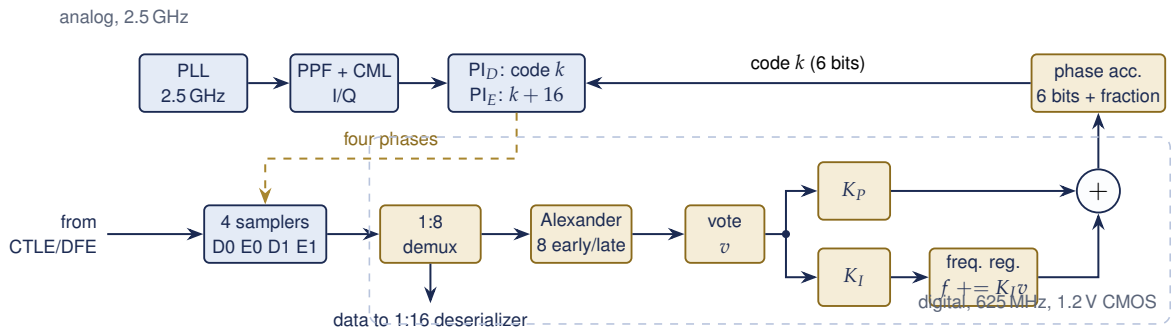
analog, 2.5 GHz
PLL
2.5 GHz
PPF + CML
I/Q
PI$_D$: code k
PI$_E$: $k + 16$
code k (6 bits)
phase acc.
6 bits + fraction
four phases
from
CTLE/DFE
4 samplers
D0 E0 D1 E1
1:8
demux
Alexander
8 early/late
vote
v
K_P
K_I
freq. reg.
$f \mathrel{+}= K_I v$
data to 1:16 deserializer
digital, 625 MHz, 1.2 V CMOS